

The Benchmark Is Part of the Option: First-Passage Prices and Sparse-Print Robustness in Compute Derivatives

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Abstract

Compute derivatives have arrived before their cash benchmark has a long public history. This paper shows that the settlement functional is therefore a first-order part of the contract, not a back-office detail. We study public event contracts on transaction-based GPU rental indices. First, a specification-aware program reproduces all 208 eligible archived strike settlements across 56 GPU-events, including terminal and running-maximum contracts. Second, we exploit matched December 31 terminal buckets and one-touch thresholds for B200, H100, and H200 on the same venue. Pathwise payoff dominance implies that the one-touch price minus an upper bound on the terminal-tail price lower-bounds the state price of crossing and subsequently ending below the barrier. Across five unresolved thresholds, the normalized lower bound is 37.9% to 45.8% at the frozen date (median 42.4%) and remains positive in every one of at least 742 aligned hourly observations per surface. Third, we define a directional sparse-print robustness radius: the smallest perturbation to one published index observation that changes a binary payoff. In an event-weighted counterfactual design, a 5% upward change to one print can change 82.4% of no-strike decisions under one-touch settlement, versus 13.2% under a monthly average; downward vulnerability is 11.1% versus 6.7%. These are deterministic susceptibility measures, not estimates of actual benchmark error or manipulation. The results motivate publication-vintage controls, averaging, and explicit path-functional disclosure as compute futures move from announced products to traded risk-transfer instruments.

Keywords: AI compute, GPU rental index, cash settlement, barrier option, prediction market, benchmark robustness.

JEL: G13, G14, G18, L86.

1 Introduction

The financialization of AI compute poses a measurement problem before it poses a forecasting problem. A rented GPU-hour is a non-storable service embedded in hardware, power, networking, software, location, and service quality. Public benchmarks compress this heterogeneous market into dollars per GPU-hour. Cash-settled con-

tracts then transform that published statistic into a payoff.

That transformation can be terminal, averaged, or path-dependent. A terminal digital asks where an index is on one date. An arithmetic-average contract asks where it was on average. A one-touch contract pays if the published index ever crossed a barrier. The same index path can settle these claims differently even when every observation is correct. When one observation is stale, revised, or otherwise disputed, the settlement rule also determines how that uncertainty enters the payoff. For a one-touch claim, a high print can be absorbing: once the contract resolves, later values cannot reverse the payout.

This issue is timely. CME announced H100 and B200 rental-index futures for an October 2026 launch, subject to regulatory review (CME Group, 2026). The CFTC’s request for comment asks directly about cash-market fragmentation, benchmark auditability, fungibility, and manipulation risk (CFTC, 2026). Meanwhile, event venues already list terminal and “hit by date” contracts on public GPU rental indices. They provide the first observable price surface on different functionals of the same compute benchmark.

We make three contributions. First, we build a settlement audit that reads the contract date from the specification rather than the UTC close timestamp, uses the stated strict threshold convention, respects the market-open window, and gates all analysis on exact reproduction. It reproduces 208 strike decisions representing 56 events. Reporting both counts matters: ladder strikes are correlated payoff decisions, not independent outcomes.

Second, we derive and measure a model-free cross-payoff bound. At a common barrier, the event that the terminal index exceeds the barrier is a subset of the event that the running maximum exceeds it. Under any non-negative state-price density, the one-touch claim is therefore worth at least the terminal digital, and their price difference is exactly the state price of crossing and reversal. When the terminal market reports coarse buckets, the full price of a bucket overlapping the barrier is a conservative upper bound on the terminal tail. In five matched active surfaces, the resulting lower bound is economically large and persistently positive. This is an implied path-state price, not a bootstrap forecast and not an objective probability.

Third, we introduce a settlement robustness object that does not require the latent economic price or the measurement-error distribution to be observed. For a payoff rule and a published path, the sparse-print radius is the smallest bounded perturbation to one print that changes the payoff. Directional radii separate false-positive and false-negative susceptibility. We then resettle the same archived event paths and listed strikes under terminal, five-day mean, seven-day median, monthly-mean, and one-touch rules. Equal event weights prevent dense ladders from dominating. The resulting frontier quantifies a design tradeoff without claiming that any perturbation actually occurred.

The paper is intentionally narrower than a compute risk-premium or latent-price study. Ninety-two public daily levels per GPU are insufficient to distinguish benchmark construction from genuine market persistence. Historical bootstrap frequencies would be physical scenario probabilities, not risk-neutral option values. We therefore center claims that are pathwise, auditable, and falsifiable with a prospective holdout.

2 Related literature

Compute as an asset-pricing object. Bandi and Su (2026) establish the key economic analogy: rented compute is a non-storable service, closer to electricity or freight capacity than to stored oil. They use term rentals to construct synthetic forward prices and study early risk-premium implications. Our object differs. We study event claims that already trade on terminal and running-maximum functionals of an Ornn index, not the term-rental curve or a synthetic futures return. Work on cloud spot pricing primarily studies allocation mechanisms, interruption, and provider pricing (Li et al., 2017); hedonic cloud-index research instead addresses quality change (Byrne et al., 2018). Neither literature identifies the path payoff priced here.

Benchmarks and cash settlement. Index construction for cash settlement is a longstanding problem (Shiller, 1993). Pirrong (2001) shows that cash settlement does not uniformly eliminate manipulation, while Duffie and Dworczak (2021) derives robust benchmark fixings when contributors have positions linked to the fixing. IOSCO principles emphasize methodology transparency, input quality, change control, and accountability (IOSCO, 2013). We do not estimate manipulation or design an optimal contributor-weighting rule. Our robustness radius takes the published path as given and asks how a sparse perturbation propagates through alternative settlement functionals.

Barrier options and prediction markets. Barrier option value depends on monitoring frequency and first passage (Broadie et al., 1997). The classical literature

typically treats the underlying path as observed under a specified stochastic process. We instead keep the path nonparametric and perturb its published observations. Dai et al. (2026) document settlement manipulation in short-horizon crypto prediction markets using spot order flow and reversals. Our data do not contain a constituent transaction tape and cannot support that test. We report susceptibility, not realized manipulation. Prediction-market prices are useful state-price signals, but interpretation requires attention to liquidity, collateral, and trading costs (Wolfer and Zitzewitz, 2004).

3 Payoff geometry and robustness

Let $Y_t > 0$ be the published daily index from a contract’s opening through expiry T , and let $K > 0$ be a high barrier. Define terminal and one-touch digitals

$$H_T(K; Y) = \mathbb{I}\{Y_T \geq K\}, \quad (1)$$

$$H_M(K; Y) = \mathbb{I}\{\max_{0 \leq t \leq T} Y_t \geq K\}. \quad (2)$$

The weak inequality matches the Polymarket rules, which state “equal to or beyond.” The Kalshi audit below uses its stated strict “above” rule. With a rounded index, this distinction is not innocuous; we preserve each venue’s text.

3.1 A model-free first-passage bound

Proposition 1 (payoff dominance). For every realized path Y and barrier K , $H_M(K; Y) \geq H_T(K; Y)$. If $m \geq 0$ is any state-price density and $\pi(H) = \mathbb{E}[mH]$, then

$$\pi(H_M) - \pi(H_T) = \mathbb{E}\left[m \mathbb{I}\left\{\max_t Y_t \geq K, Y_T < K\right\}\right] \geq 0. \quad (3)$$

Proof. If $Y_T \geq K$, the terminal observation is itself in the running maximum, so $\max_t Y_t \geq K$. Subtracting the two indicators leaves exactly the crossing-and-reversal state. Multiplication by nonnegative m and expectation preserve the inequality. $\square$

Suppose a terminal market lists mutually exclusive buckets with prices q_j . We divide by $\sum_j q_j$ to remove the observed event-level overround. If K falls inside a bucket rather than on a boundary, assigning the entire overlapping bucket to the upper tail gives $U_T(K) \geq \pi(H_T(K))$. Thus

$$L(K) = \max\{0, p_M(K) - U_T(K)\} \quad (4)$$

is a conservative lower bound on the price-implied crossing-and-reversal state, conditional on a common pricing measure. Raw, unnormalized bounds are retained as a robustness check.

This is not automatically an executable arbitrage result. The archive lacks historical bids, asks, sizes, and gas or fee data; separate event pools can also have different marginal traders. Equation (4) is a payoff-coherence diagnostic and state-price interpretation.

3.2 Directional sparse-print radius

Let $\phi_r(Y)$ be a settlement functional: terminal, arithmetic mean, median, or maximum. For a binary claim $H_{r,K}(Y) = \mathbb{I}\{\phi_r(Y) > K\}$, define the one-print log robustness radius

$$\rho_{r,K}^s(Y) = \inf_{j, \epsilon \geq 0} \left\{ \epsilon : H_{r,K}(Y_1, \dots, Y_j e^{s\epsilon}, \dots, Y_T) \neq H_{r,K}(Y) \right\}, \quad (5)$$

where $s = +1$ is an upward perturbation and $s = -1$ a downward perturbation. The radius is infinite when no one-print perturbation in that direction can change the payoff.

The direction has a settlement interpretation. For a base “No,” an upward flip is false-positive susceptibility. For a base “Yes,” a downward flip is false-negative susceptibility. A mean attenuates a one-print level error by its weight. A terminal claim gives only the final print nonzero influence. A maximum has every monitored date as a possible positive trigger. Conversely, after multiple valid crossings, no one downward change can reverse a touch payout. This produces asymmetric robustness even without a stochastic noise model.

4 Data and reproducibility

The package freezes three inputs. First, five Ornn GPU index paths contain 92 daily levels each from May 30 through August 29, 2026. Public Ornn materials describe the current compute index as a transaction-based, volume-weighted, winsorized mean (Ornn, 2026). We treat that description as a vendor claim, not independently verified transaction provenance.

Second, 17 Kalshi compute series contain contract rules, strikes, opening and closing timestamps, results, and settlement values. After requiring a complete index window and parseable contract date, the sample contains 208 strike decisions across 56 events: 179 terminal-rule decisions and 29 running-maximum decisions. The latter occupy only three event windows. All inference and sensitivity summaries therefore weight events equally; strike-weighted counts are secondary diagnostics.

Third, six Polymarket events pair terminal and “hit by December 31” markets for B200, H100, and H200. The terminal contracts settle to the finalized December 31 Ornn daily value. The touch contracts settle as soon as a daily index is equal to or beyond the barrier. Their rules also state that a previously published crossing is not disqualified by a subsequent revision (Polymarket, 2026a,b). We retain YES-token histories, event liquidity, and snapshot outcome prices. Five unresolved high barriers have a usable matched terminal tail.

The frozen source manifest stores SHA-256 hashes of parent snapshots and compact study inputs. Analysis verifies those hashes before calculation. No target fetches live data. The code, tests, machine-readable outputs, and

manuscript are generated under `novelty/`. This separation matters because the parent repository contained derived artifacts from multiple retrieval times; none is used in the paper.

4.1 Settlement reproduction gate

Dates are parsed from event tickers, not UTC close timestamps: a contract that settles at 11:59 p.m. Eastern can carry a next-day UTC timestamp. Touch windows begin at market opening or the first archived index date, whichever is later. The audit program computes the named functional and compares both the binary result and numeric settlement value.

All 208 binary outcomes reproduce. All 187 contracts with numeric settlement values match the computed functional within half a cent. This gate does not prove the upstream index was correct at publication time; it proves that the archived contract record and archived index path are mutually consistent under the encoded specification.

5 Matched terminal and touch surfaces

Table 1 reports the five active high barriers. B200’s terminal market has a single open-ended \$7+ bucket, so the full bucket price is an upper bound for both \$7.50 and \$8.00 terminal tails. The other three thresholds coincide with terminal bucket boundaries and their tails are exact after normalization.

The cross-payoff wedge is not a small correction. Snapshot lower bounds range from 37.9% to 45.8%, with mean 42.1%. The sign is not a last-hour artifact: each of the five bounds is positive in every one of 742 or 743 common hourly observations. The smallest within-series historical minimum is 7.2%; series medians range from 24.2% to 46.1%.

Economically, the result says that the market places material value on paths that cross a high rental-price threshold and then finish below it. That state can reflect genuine temporary scarcity, mean reversion, benchmark publication dynamics, or different liquidity and risk premia across event pools. The design does not identify which mechanism is responsible. It does reject the short-cut of reading a “hit by date” price as a terminal futures expectation.

6 Sparse-print robustness frontier

We next re-settle each eligible path and listed strike under five functionals: terminal, trailing five-day mean, trailing seven-day median, calendar-month mean, and running maximum. For each rule, outcome, and direction, we solve Equation (5) by bisection over multiplicative changes up to 100x. At a perturbation budget ϵ , a strike

Table 1: Matched one-touch and terminal-tail prices on August 29, 2026. Prices are dollars per \$1 payoff. “Bound” is the normalized lower bound from Equation (4). History columns summarize the same bound over aligned hourly observations. B200 terminal tails are conservative upper bounds because the barrier lies inside the \$7+ bucket.

GPU	K	Touch	Terminal tail	Bound	Hours	Hist. min	Hist. median	Positive
B200	8.00	.475	.085	.390	743	.072	.242	100%
B200	7.50	.510	.085	.424	743	.299	.434	100%
H100	3.50	.630	.172	.458	742	.260	.452	100%
H200	7.00	.417	.038	.379	742	.152	.363	100%
H200	6.00	.600	.144	.456	742	.142	.461	100%

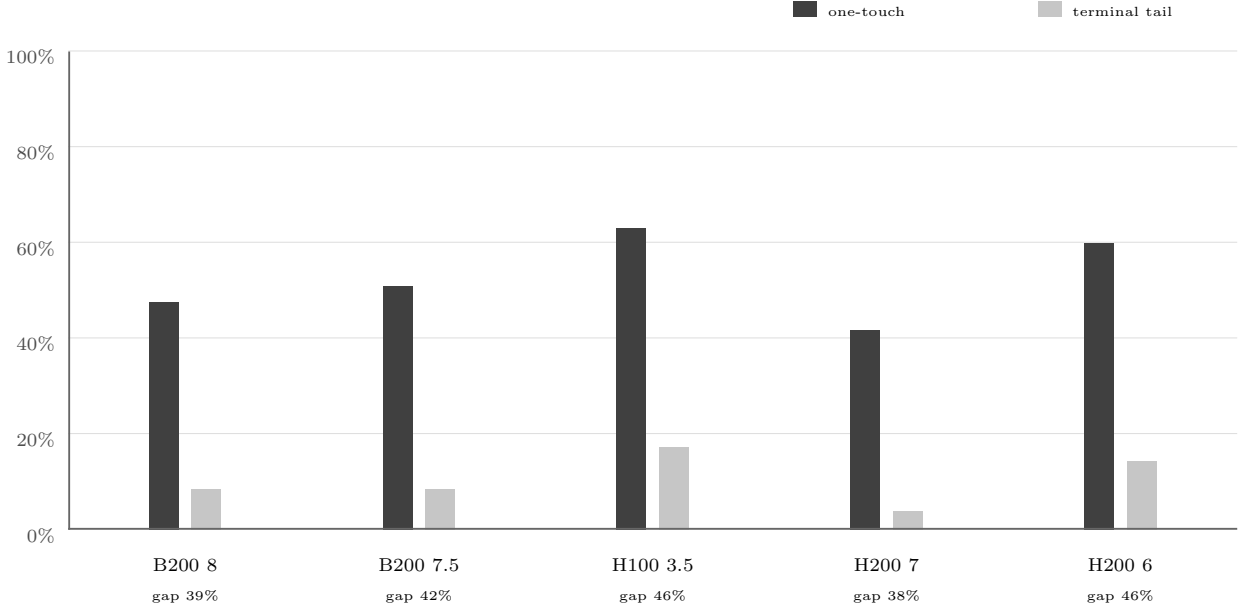


Figure 1: One-touch prices and normalized terminal-tail upper bounds. The gap is a lower bound on the price-implied crossing-and-reversal state under a common pricing measure. It is not an objective probability or an executable arbitrage return.

Table 2: Event-weighted share of listed strike decisions vulnerable to one 5% print perturbation under counterfactual settlement rules. Upward perturbations are evaluated for base-No outcomes; downward perturbations for base-Yes.

Rule	Up / FP	Down / FN
Terminal	54.6%	53.0%
Mean, 5 days	12.3%	18.3%
Median, 7 days	15.3%	22.8%
Calendar-month mean	13.2%	6.7%
One-touch maximum	82.4%	11.1%

is vulnerable if $\rho \leq \log(1 + \epsilon)$. We average vulnerable-strike shares within each event and then average events. This estimand gives a one-strike event and a 20-strike ladder the same total weight.

The asymmetry is sharp. A one-touch rule exposes every monitored print as a potential high trigger, so 82.4% of base-No decisions are vulnerable to some single 5% upward change. Yet only 11.1% of base-Yes decisions can

be reversed by one downward change. A monthly average reduces those shares to 13.2% and 6.7%. The counterfactual is not a causal estimate of a venue redesign: changing the functional also changes rule-specific moneyness. It is a deterministic answer to a narrower question: on these paths and listed strikes, which payouts can one changed print alter?

Two robustness checks limit the role of unusual products. Excluding A100 and RTX 5090 leaves only B200, H100, and H200. The 5% upward event-weighted share is still 75.8% for touch versus 11.1% for monthly mean. Restricting each event to its actual settlement rule preserves the exact audit but leaves only three touch events, so we do not attach a sampling p-value to that comparison. Full curves from 0.5% through 10%, actual-rule-only results, all directional radii, and strike-weighted secondary summaries are included in the machine-readable outputs.

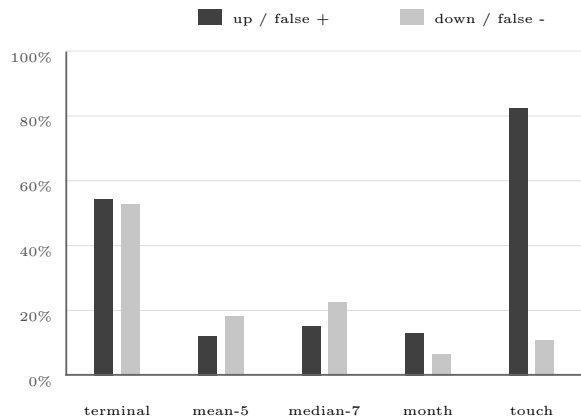


Figure 2: The same 5% sparse-print experiment. A one-touch maximum is especially susceptible to upward changes near a listed barrier but comparatively robust to one downward change after one or more crossings.

7 Implications for contract and benchmark design

Name the path functional. “Above on date” and “above by date” are different options. Exchange feeds, risk systems, and disclosure should encode the functional as a first-class field rather than leave it in prose. A terminal digital cannot be substituted for a one-touch in calibration or hedge accounting.

Version the benchmark. A touch rule that preserves a crossing even after the source revises that print is a claim on a publication vintage, not merely a final time series. Contract records therefore need observation time, publication time, retrieval time, methodology version, and revision policy. A final revised history is insufficient for an independent audit.

Choose aggregation for the exposure. If the hedged cash flow is average monthly compute expenditure, an averaged settlement is economically aligned and attenuates isolated print influence. If the exposure is a capacity covenant triggered by any scarcity spike, a maximum may be appropriate, but its asymmetric robustness and monitoring window should be priced explicitly. No functional dominates for every exposure.

Publish contributor and change controls. The sparse radius is only a payoff-level diagnostic. It complements, rather than replaces, robust contributor weighting, transaction validation, break registers, governance, and audit controls emphasized by IOSCO and benchmark- design research. Regulators should ask both how difficult the index is to move and how a given move propagates into payouts.

8 Limitations and prospective tests

The study is a discovery sample. Five matched active thresholds are enough to establish an empirical object, not a universal law of compute markets. Hourly observations are highly dependent and are presented as a persistence audit, not 742 independent replications. The archive lacks historical order books, so the state-price wedge is not a net trading return. Separate event pools can carry different risk premia and funding constraints. Normalization removes event- level overround but not these frictions.

The robustness frontier is conditional on observed paths and listed strikes. It does not estimate the frequency, sign, or magnitude of actual benchmark errors. A genuine market move and a measurement error are observationally distinct concepts, and daily return volatility cannot identify the latter. The audit uses later-retrieved history; silent revisions between original settlement and retrieval remain possible.

We predeclare four prospective falsification tests for observations first published after August 30, 2026:

1. Repeat the matched-surface bound with time-stamped executable bids, asks, sizes, fees, and a common collateral clock. The price-coherence interpretation fails if the lower bounds vanish after executable costs.
2. Apply the frozen settlement parser to new events before outcomes are known. Any rule-dependent reproduction failure invalidates the current taxonomy.
3. Publish the event-weighted sparse frontier on a forward holdout without changing the perturbation grid, rule set, or strict/weak inequality convention. The design result weakens if touch false-positive susceptibility is no larger than averaging at comparable listed moneyness.
4. Add point-in-time benchmark vintages. If crossings routinely disappear before finalization and contracts correctly ignore them, the claimed absorbing publication-vintage channel is rejected.

Hedge effectiveness is deliberately outside the current claims. It requires participant cash-price and usage data, not just the reference index. Likewise, no risk premium can be identified from these binary prices without an explicit pricing-kernel model and substantially more independent settlements.

9 Conclusion

Compute is becoming tradable through indices before those indices have the history and governance of mature commodity benchmarks. In that setting, the benchmark

is part of the option, and the settlement functional determines which features of its path have economic value.

The paper supplies three auditable results: exact reproduction of 208 archived strike settlements; a model-free and persistently positive first-passage price bound across matched terminal/touch surfaces; and an event-weighted sparse-print robustness frontier showing the directional asymmetry of one-touch settlement. None requires relabeling index volatility as measurement error or a historical bootstrap as a fair option price.

For market designers, the practical message is direct. Specify the path functional, preserve publication vintages, align averaging with the exposure, and disclose how revisions affect already-triggered claims. For researchers, the next milestone is a prospective, executable-price holdout. That evidence can turn the present robustness audit into a test of pricing and market quality as listed compute futures begin trading.

Reproducibility statement

The frozen data, SHA-256 manifest, standard-library Python analysis, unit tests, CSV and JSON results, vector figures, and this manuscript are included in the artifact. Running `make -C novelty all` performs no network access. The primary numerical claims are generated into the manuscript from `paper/generated/results_macros.tex`. Source limitations and rejected interpretations are recorded in `DATA_CARD.md` and `literature/RESEARCH_LOG.md`.

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